Results for algebra2005

original held-out split; N = 5, 10, ..., 50; model seeds 2024, 2025, 2026.

Six metrics: Ada, NOV, and Ep at response thresholds 0.4, 0.5, 0.6, 0.7.

Mean ± sample SD across available model seeds; a lone run is shown without SD.

Bold = best; underline = second best by unrounded mean, among completed methods.

Ablation models	Seeds
2CKG4ER	3/3
No-mastery	3/3
No-forgetting	3/3
No-student-feature	3/3
No-exercise-feature	3/3
ID-based only	3/3

Comparison models	Seeds
EB-CF	fixed
SB-CF	fixed
CBF	fixed
TransE	3/3
TransE-adv	3/3
RotatE	1/3
DistMult	0/3
ComplEx	0/3
2CKG4ER	3/3

Interpretation of incomplete results

A dash means no evaluated result was in this archive. Incomplete methods are shown but excluded from ranking.

CF baselines are deterministic and displayed as one fixed value. Only 3/3 seeded methods enter ranking.

Source archive: Algebra2005_current_results_20261010_223303.zip

Ablation performance

Six variants; all 18 model/seed evaluations available.

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.790 ± 0.001**	0.405 ± 0.041	0.699 ± 0.000	0.723 ± 0.010	0.773 ± 0.004	0.683 ± 0.025
10	0.743 ± 0.005	0.425 ± 0.033	0.699 ± 0.000	0.699 ± 0.018	**0.745 ± 0.003**	0.674 ± 0.016
15	0.716 ± 0.020	0.426 ± 0.034	0.699 ± 0.000	0.681 ± 0.021	**0.721 ± 0.012**	0.664 ± 0.011
20	**0.712 ± 0.017**	0.410 ± 0.015	0.700 ± 0.002	0.669 ± 0.025	0.711 ± 0.009	0.652 ± 0.006
25	0.692 ± 0.004	0.410 ± 0.015	**0.703 ± 0.003**	0.656 ± 0.025	0.692 ± 0.006	0.642 ± 0.004
30	0.673 ± 0.006	0.414 ± 0.013	**0.708 ± 0.003**	0.645 ± 0.023	0.675 ± 0.007	0.631 ± 0.004
35	0.662 ± 0.010	0.409 ± 0.017	**0.711 ± 0.003**	0.634 ± 0.023	0.665 ± 0.004	0.623 ± 0.004
40	0.655 ± 0.006	0.400 ± 0.019	**0.715 ± 0.001**	0.624 ± 0.024	0.658 ± 0.003	0.615 ± 0.006
45	0.649 ± 0.004	0.392 ± 0.021	**0.715 ± 0.001**	0.613 ± 0.023	0.653 ± 0.002	0.607 ± 0.008
50	0.644 ± 0.007	0.386 ± 0.023	**0.714 ± 0.002**	0.603 ± 0.022	0.649 ± 0.003	0.599 ± 0.008

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.993 ± 0.001**	0.975 ± 0.002	0.991 ± 0.000	0.990 ± 0.001	0.991 ± 0.002	0.986 ± 0.006
10	0.988 ± 0.001	0.975 ± 0.001	**0.991 ± 0.000**	0.989 ± 0.003	0.986 ± 0.001	0.985 ± 0.004
15	0.985 ± 0.001	0.975 ± 0.001	**0.991 ± 0.000**	0.988 ± 0.003	0.984 ± 0.001	0.985 ± 0.004
20	0.984 ± 0.000	0.974 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.004	0.984 ± 0.001	0.985 ± 0.004
25	0.985 ± 0.000	0.974 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.003	0.984 ± 0.000	0.986 ± 0.003
30	0.985 ± 0.001	0.974 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.003	0.984 ± 0.000	0.985 ± 0.003
35	0.985 ± 0.001	0.974 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.004	0.984 ± 0.000	0.986 ± 0.003
40	0.985 ± 0.001	0.973 ± 0.002	**0.991 ± 0.000**	0.987 ± 0.003	0.985 ± 0.000	0.986 ± 0.003
45	0.985 ± 0.000	0.973 ± 0.001	**0.991 ± 0.000**	0.987 ± 0.003	0.985 ± 0.000	0.986 ± 0.002
50	0.985 ± 0.000	0.973 ± 0.001	**0.990 ± 0.001**	0.987 ± 0.002	0.985 ± 0.000	0.986 ± 0.002

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Ablation performance

Six variants; all 18 model/seed evaluations available.

(c) Ep, threshold 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.162 ± 0.028	0.048 ± 0.086	**0.176 ± 0.005**	-0.044 ± 0.018	-0.110 ± 0.042	0.014 ± 0.035
10	0.001 ± 0.034	0.089 ± 0.021	**0.228 ± 0.006**	0.007 ± 0.019	0.036 ± 0.025	0.042 ± 0.025
15	-0.008 ± 0.154	0.120 ± 0.031	**0.253 ± 0.009**	0.055 ± 0.008	0.059 ± 0.049	0.091 ± 0.020
20	0.042 ± 0.041	0.092 ± 0.018	**0.237 ± 0.032**	0.088 ± 0.005	0.039 ± 0.034	0.087 ± 0.002
25	-0.063 ± 0.067	0.116 ± 0.031	**0.169 ± 0.094**	0.111 ± 0.011	-0.025 ± 0.030	0.092 ± 0.006
30	0.007 ± 0.048	0.080 ± 0.019	0.027 ± 0.134	0.112 ± 0.011	0.075 ± 0.103	**0.115 ± 0.008**
35	**0.178 ± 0.065**	0.093 ± 0.012	0.025 ± 0.042	0.123 ± 0.008	0.092 ± 0.068	0.117 ± 0.032
40	**0.209 ± 0.021**	0.039 ± 0.041	-0.011 ± 0.044	0.133 ± 0.009	0.156 ± 0.056	0.125 ± 0.022
45	0.205 ± 0.034	0.050 ± 0.056	0.110 ± 0.017	0.135 ± 0.011	**0.211 ± 0.005**	0.135 ± 0.024
50	0.197 ± 0.045	0.035 ± 0.049	0.147 ± 0.055	0.125 ± 0.008	**0.214 ± 0.012**	0.132 ± 0.025

(d) Ep, threshold 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.169 ± 0.025	-0.007 ± 0.127	**0.172 ± 0.005**	-0.061 ± 0.020	-0.154 ± 0.078	-0.005 ± 0.043
10	-0.009 ± 0.040	0.063 ± 0.009	**0.227 ± 0.006**	-0.038 ± 0.032	0.005 ± 0.021	0.015 ± 0.031
15	-0.018 ± 0.167	0.108 ± 0.029	**0.252 ± 0.009**	0.012 ± 0.022	0.052 ± 0.050	0.068 ± 0.025
20	0.025 ± 0.039	0.080 ± 0.023	**0.237 ± 0.032**	0.060 ± 0.012	0.037 ± 0.034	0.068 ± 0.003
25	-0.066 ± 0.064	0.102 ± 0.040	**0.169 ± 0.094**	0.088 ± 0.013	-0.043 ± 0.036	0.075 ± 0.006
30	-0.006 ± 0.065	0.065 ± 0.026	0.026 ± 0.133	0.092 ± 0.013	0.063 ± 0.099	**0.096 ± 0.005**
35	**0.171 ± 0.075**	0.075 ± 0.017	0.020 ± 0.048	0.103 ± 0.010	0.088 ± 0.069	0.099 ± 0.033
40	**0.205 ± 0.027**	0.019 ± 0.040	-0.033 ± 0.074	0.114 ± 0.015	0.152 ± 0.054	0.104 ± 0.023
45	0.202 ± 0.031	0.038 ± 0.051	0.106 ± 0.013	0.112 ± 0.015	**0.209 ± 0.006**	0.115 ± 0.028
50	0.194 ± 0.042	0.025 ± 0.050	0.147 ± 0.056	0.098 ± 0.014	**0.212 ± 0.013**	0.107 ± 0.027

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Ablation performance

Six variants; all 18 model/seed evaluations available.

(e) Ep, threshold 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.299 ± 0.010	-0.071 ± 0.169	**0.112 ± 0.002**	-0.147 ± 0.030	-0.284 ± 0.065	-0.075 ± 0.052
10	-0.173 ± 0.059	-0.011 ± 0.010	**0.170 ± 0.009**	-0.164 ± 0.041	-0.191 ± 0.025	-0.091 ± 0.044
15	-0.150 ± 0.238	0.042 ± 0.051	**0.200 ± 0.006**	-0.136 ± 0.038	-0.086 ± 0.093	-0.056 ± 0.035
20	-0.087 ± 0.132	0.001 ± 0.029	**0.192 ± 0.039**	-0.101 ± 0.035	-0.100 ± 0.108	-0.065 ± 0.013
25	-0.284 ± 0.030	0.048 ± 0.064	**0.125 ± 0.086**	-0.070 ± 0.014	-0.238 ± 0.039	-0.062 ± 0.007
30	-0.260 ± 0.065	**0.010 ± 0.035**	-0.183 ± 0.125	-0.061 ± 0.037	-0.155 ± 0.141	-0.032 ± 0.018
35	-0.016 ± 0.179	**0.018 ± 0.030**	-0.305 ± 0.095	-0.048 ± 0.026	-0.095 ± 0.025	-0.029 ± 0.046
40	**0.102 ± 0.053**	-0.033 ± 0.042	-0.404 ± 0.008	-0.034 ± 0.030	0.015 ± 0.060	-0.033 ± 0.045
45	**0.126 ± 0.017**	-0.013 ± 0.064	-0.321 ± 0.069	-0.024 ± 0.030	0.120 ± 0.008	-0.018 ± 0.055
50	0.130 ± 0.041	-0.029 ± 0.075	-0.269 ± 0.074	-0.038 ± 0.022	**0.148 ± 0.006**	-0.019 ± 0.055

(f) Ep, threshold 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.363 ± 0.013	-0.130 ± 0.155	**-0.008 ± 0.006**	-0.222 ± 0.037	-0.366 ± 0.052	-0.172 ± 0.055
10	-0.304 ± 0.055	-0.153 ± 0.042	**-0.003 ± 0.010**	-0.269 ± 0.045	-0.289 ± 0.025	-0.224 ± 0.037
15	-0.313 ± 0.143	-0.086 ± 0.051	**0.002 ± 0.006**	-0.256 ± 0.031	-0.273 ± 0.047	-0.220 ± 0.040
20	-0.330 ± 0.032	-0.146 ± 0.021	**-0.056 ± 0.080**	-0.240 ± 0.029	-0.303 ± 0.087	-0.235 ± 0.029
25	-0.429 ± 0.032	**-0.061 ± 0.088**	-0.152 ± 0.101	-0.230 ± 0.009	-0.406 ± 0.029	-0.246 ± 0.016
30	-0.373 ± 0.061	**-0.127 ± 0.057**	-0.372 ± 0.108	-0.234 ± 0.021	-0.309 ± 0.114	-0.230 ± 0.017
35	-0.136 ± 0.172	**-0.131 ± 0.047**	-0.400 ± 0.033	-0.230 ± 0.021	-0.293 ± 0.031	-0.234 ± 0.042
40	**-0.044 ± 0.127**	-0.195 ± 0.048	-0.428 ± 0.019	-0.228 ± 0.017	-0.240 ± 0.067	-0.234 ± 0.042
45	**-0.016 ± 0.042**	-0.181 ± 0.062	-0.335 ± 0.054	-0.223 ± 0.023	-0.149 ± 0.049	-0.230 ± 0.045
50	**-0.039 ± 0.130**	-0.194 ± 0.074	-0.309 ± 0.035	-0.229 ± 0.009	-0.111 ± 0.108	-0.229 ± 0.044

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines plus 2CKG4ER; missing runs are shown as dashes.

(a) Adaptability (Ada)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.222	0.450	0.274	0.730 ± 0.005	0.725 ± 0.005	0.787	-	-	**0.790 ± 0.001**
10	0.220	0.515	0.269	0.701 ± 0.005	0.711 ± 0.008	0.758	-	-	**0.743 ± 0.005**
15	0.219	0.498	0.269	0.672 ± 0.005	0.689 ± 0.004	0.720	-	-	**0.716 ± 0.020**
20	0.217	0.468	0.271	0.653 ± 0.006	0.676 ± 0.006	0.693	-	-	**0.712 ± 0.017**
25	0.218	0.441	0.272	0.637 ± 0.008	0.668 ± 0.008	0.671	-	-	**0.692 ± 0.004**
30	0.217	0.418	0.272	0.624 ± 0.009	0.659 ± 0.008	0.658	-	-	**0.673 ± 0.006**
35	0.217	0.398	0.273	0.613 ± 0.008	0.654 ± 0.009	0.647	-	-	**0.662 ± 0.010**
40	0.216	0.381	0.275	0.604 ± 0.009	0.649 ± 0.010	0.641	-	-	**0.655 ± 0.006**
45	0.216	0.366	0.275	0.594 ± 0.009	0.642 ± 0.010	0.629	-	-	**0.649 ± 0.004**
50	0.216	0.355	0.274	0.586 ± 0.009	0.633 ± 0.010	0.618	-	-	**0.644 ± 0.007**

(b) Novelty (NOV)

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	0.957	0.975	0.993	0.991 ± 0.001	**0.996 ± 0.001**	0.993	-	-	0.993 ± 0.001
10	0.961	0.979	0.994	0.992 ± 0.000	**0.994 ± 0.000**	0.990	-	-	0.988 ± 0.001
15	0.964	0.982	**0.995**	0.994 ± 0.000	0.993 ± 0.001	0.988	-	-	0.985 ± 0.001
20	0.964	0.981	0.994	**0.995 ± 0.000**	0.994 ± 0.000	0.989	-	-	0.984 ± 0.000
25	0.965	0.981	0.991	**0.994 ± 0.000**	0.993 ± 0.001	0.989	-	-	0.985 ± 0.000
30	0.966	0.981	0.990	**0.993 ± 0.000**	0.993 ± 0.001	0.989	-	-	0.985 ± 0.001
35	0.965	0.980	0.990	**0.993 ± 0.000**	0.992 ± 0.000	0.989	-	-	0.985 ± 0.001
40	0.965	0.979	0.990	**0.993 ± 0.000**	0.992 ± 0.001	0.990	-	-	0.985 ± 0.001
45	0.965	0.980	0.990	**0.993 ± 0.000**	0.991 ± 0.001	0.990	-	-	0.985 ± 0.000
50	0.964	0.980	0.990	**0.993 ± 0.000**	0.991 ± 0.000	0.990	-	-	0.985 ± 0.000

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines plus 2CKG4ER; missing runs are shown as dashes.

(c) Ep, threshold 0.4

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.037	0.047	**0.136**	-0.102 ± 0.012	-0.054 ± 0.018	-0.162	-	-	-0.162 ± 0.028
10	0.021	**0.163**	0.157	-0.008 ± 0.004	0.009 ± 0.015	-0.056	-	-	0.001 ± 0.034
15	0.002	**0.188**	0.164	-0.003 ± 0.015	-0.018 ± 0.023	0.035	-	-	-0.008 ± 0.154
20	-0.019	**0.164**	0.154	0.026 ± 0.023	0.058 ± 0.027	0.048	-	-	0.042 ± 0.041
25	-0.018	**0.171**	0.164	0.099 ± 0.010	0.101 ± 0.017	0.005	-	-	-0.063 ± 0.067
30	-0.032	**0.167**	0.146	0.144 ± 0.025	0.137 ± 0.005	0.108	-	-	0.007 ± 0.048
35	-0.021	0.150	0.155	**0.189 ± 0.004**	0.145 ± 0.012	0.166	-	-	0.178 ± 0.065
40	-0.045	0.147	0.141	0.185 ± 0.003	0.154 ± 0.019	0.168	-	-	**0.209 ± 0.021**
45	-0.015	0.146	0.139	0.169 ± 0.011	0.163 ± 0.014	0.186	-	-	**0.205 ± 0.034**
50	-0.026	0.154	0.128	0.160 ± 0.044	0.097 ± 0.019	0.177	-	-	**0.197 ± 0.045**

(d) Ep, threshold 0.5

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.039	0.041	**0.101**	-0.116 ± 0.011	-0.061 ± 0.017	-0.175	-	-	-0.169 ± 0.025
10	0.019	**0.159**	0.121	-0.041 ± 0.015	-0.010 ± 0.017	-0.186	-	-	-0.009 ± 0.040
15	-0.001	**0.183**	0.136	-0.045 ± 0.022	-0.042 ± 0.029	-0.058	-	-	-0.018 ± 0.167
20	-0.021	**0.156**	0.130	-0.014 ± 0.016	0.026 ± 0.030	0.009	-	-	0.025 ± 0.039
25	-0.021	**0.162**	0.136	0.063 ± 0.015	0.075 ± 0.021	-0.012	-	-	-0.066 ± 0.064
30	-0.032	**0.161**	0.120	0.114 ± 0.033	0.119 ± 0.002	0.094	-	-	-0.006 ± 0.065
35	-0.024	0.136	0.127	0.167 ± 0.009	0.133 ± 0.011	0.154	-	-	**0.171 ± 0.075**
40	-0.045	0.137	0.109	0.165 ± 0.004	0.140 ± 0.017	0.156	-	-	**0.205 ± 0.027**
45	-0.017	0.139	0.102	0.147 ± 0.010	0.146 ± 0.013	0.175	-	-	**0.202 ± 0.031**
50	-0.030	0.146	0.092	0.134 ± 0.044	0.078 ± 0.021	0.168	-	-	**0.194 ± 0.042**

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.

Comparison performance

Eight baselines plus 2CKG4ER; missing runs are shown as dashes.

(e) Ep, threshold 0.6

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.042	0.002	**0.045**	-0.209 ± 0.008	-0.156 ± 0.017	-0.307	-	-	-0.299 ± 0.010
10	0.011	**0.082**	0.049	-0.198 ± 0.009	-0.134 ± 0.011	-0.365	-	-	-0.173 ± 0.059
15	-0.011	**0.135**	0.061	-0.248 ± 0.020	-0.217 ± 0.025	-0.282	-	-	-0.150 ± 0.238
20	-0.034	**0.110**	0.067	-0.229 ± 0.015	-0.187 ± 0.038	-0.184	-	-	-0.087 ± 0.132
25	-0.032	**0.128**	0.066	-0.167 ± 0.023	-0.129 ± 0.012	-0.179	-	-	-0.284 ± 0.030
30	-0.043	**0.118**	0.046	-0.098 ± 0.041	-0.024 ± 0.019	-0.079	-	-	-0.260 ± 0.065
35	-0.034	**0.095**	0.059	-0.024 ± 0.015	0.022 ± 0.025	0.023	-	-	-0.016 ± 0.179
40	-0.060	**0.102**	0.049	0.043 ± 0.010	0.037 ± 0.041	0.054	-	-	0.102 ± 0.053
45	-0.031	0.108	0.033	0.035 ± 0.010	0.052 ± 0.032	0.097	-	-	**0.126 ± 0.017**
50	-0.046	0.115	0.022	0.035 ± 0.056	-0.019 ± 0.039	0.083	-	-	**0.130 ± 0.041**

(f) Ep, threshold 0.7

N	EB-CF	SB-CF	CBF	TransE	TransE-adv	RotatE	DistMult	ComplEx	2CKG4ER
5	-0.078	-0.101	**-0.014**	-0.286 ± 0.008	-0.241 ± 0.011	-0.364	-	-	-0.363 ± 0.013
10	-0.036	-0.066	**-0.033**	-0.268 ± 0.014	-0.244 ± 0.020	-0.411	-	-	-0.304 ± 0.055
15	-0.075	**-0.022**	-0.045	-0.316 ± 0.019	-0.310 ± 0.020	-0.337	-	-	-0.313 ± 0.143
20	-0.106	-0.046	**-0.038**	-0.293 ± 0.016	-0.279 ± 0.018	-0.312	-	-	-0.330 ± 0.032
25	-0.099	**-0.023**	-0.074	-0.253 ± 0.023	-0.250 ± 0.013	-0.333	-	-	-0.429 ± 0.032
30	-0.120	**-0.013**	-0.099	-0.185 ± 0.041	-0.171 ± 0.028	-0.234	-	-	-0.373 ± 0.061
35	-0.135	**-0.019**	-0.096	-0.127 ± 0.020	-0.174 ± 0.031	-0.170	-	-	-0.136 ± 0.172
40	-0.157	**-0.002**	-0.116	-0.079 ± 0.019	-0.168 ± 0.043	-0.176	-	-	-0.044 ± 0.127
45	-0.136	**0.010**	-0.125	-0.102 ± 0.002	-0.145 ± 0.039	-0.170	-	-	-0.016 ± 0.042
50	-0.167	**0.029**	-0.137	-0.123 ± 0.061	-0.215 ± 0.037	-0.166	-	-	-0.039 ± 0.130

(theta_m, theta_f, rec) = (0.8, 0.8, 50); zero exercise-forgetting edges removed.